

ECE4010J Definitions and Concepts Checklist

Basic Impression about Each Term

1 Probability Philosophy and Early Concepts

Problem of Points / Fair Division

- Fair division depends on **future winning probabilities**, not just the current score.
- Fermat's method: enumerate possible future outcomes and split according to winning chances.

Probability Interpretations

- **Classical probability**: based on symmetry and equally likely outcomes.
- **Frequentist probability**: long-run relative frequency in repeated identical experiments.
- **Subjective probability**: coherent degree of belief.
- **Dutch book / arbitrage**: inconsistent subjective probabilities allow a guaranteed-loss betting scheme.

Rare Events and Poisson Approximation Idea

- Law of Truly Large Numbers: with enough opportunities, rare events become likely to occur.
- If many independent opportunities have small success probabilities and expected count μ , then

$$\mathbb{P}(N = 0) \approx e^{-\mu}, \quad \mathbb{P}(N \geq 1) \approx 1 - e^{-\mu}.$$

2 Basic Probability Space

Sample Space and Events

- **Sample space**: Ω , the set of all possible outcomes.
- **Sample point**: $\omega \in \Omega$.
- **Event**: a suitable subset $A \subseteq \Omega$.
- Impossible event: $\emptyset$.
- Certain event: Ω .

Event Space / σ -algebra

- The event space is denoted by $\mathcal{F}$.
- A σ -algebra satisfies

$$\emptyset \in \mathcal{F}, \quad A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}, \quad A_i \in \mathcal{F} \Rightarrow \bigcup_{i \geq 1} A_i \in \mathcal{F}.$$

- If Ω is countable, usually take $\mathcal{F} = \mathcal{P}(\Omega)$.

Probability Space

- A **probability space** is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}).$$

- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$.
- Normalization:

$$\mathbb{P}(\Omega) = 1.$$

- Countable additivity: for mutually exclusive A_k ,

$$\mathbb{P}\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(A_k).$$

Basic Probability Rules

- $\mathbb{P}(\emptyset) = 0$.
- Complement rule:

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

- Two-event union rule:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

- Boole's inequality:

$$\mathbb{P}\left(\bigcup_{i \geq 1} A_i\right) \leq \sum_{i \geq 1} \mathbb{P}(A_i).$$

- Inclusion-exclusion:

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) &= \sum_i \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i \cap A_j) \\ &\quad + \sum_{i < j < k} \mathbb{P}(A_i \cap A_j \cap A_k) - \dots \end{aligned}$$

3 Conditional Probability and Independence

Conditional Probability

- For $\mathbb{P}(A) \neq 0$,

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}.$$

Independence of Events

- A and B are independent iff

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

- Equivalently, if $\mathbb{P}(A) \neq 0$,

$$\mathbb{P}(B \mid A) = \mathbb{P}(B).$$

- Disjoint means $A \cap B = \emptyset$.
- Independent means $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
- If $\mathbb{P}(A), \mathbb{P}(B) > 0$, disjoint events are **not** independent.

Pairwise vs Mutual Independence

- Pairwise independence:

$$\mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j), \quad i \neq j.$$

- Mutual independence: for every subcollection,

$$\mathbb{P}\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k \mathbb{P}(A_{i_j}).$$

- Pairwise independence does **not** imply mutual independence.

Partition, Total Probability, Bayes Formula

- A partition $\{A_i\}_{i \in I}$ satisfies

$$\bigcup_i A_i = \Omega, \quad A_i \cap A_j = \emptyset \quad (i \neq j).$$

- Total probability:

$$\mathbb{P}(B) = \sum_i \mathbb{P}(B \cap A_i) = \sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i).$$

- Bayes formula:

$$\mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)}{\sum_j \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)}.$$

- Two-case Bayes formula:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c)}.$$

- Odds form:

$$\frac{\mathbb{P}(A \mid B)}{\mathbb{P}(A^c \mid B)} = \frac{\mathbb{P}(A)}{\mathbb{P}(A^c)} \cdot \frac{\mathbb{P}(B \mid A)}{\mathbb{P}(B \mid A^c)}.$$

4 Counting Concepts

Basic Counting Table

Case	Meaning	Formula
Ordered, repetition allowed	sequence with replacement	n^r
Ordered, no repetition	sequence without replacement	$n(n-1) \cdots (n-r+1)$
Unordered, no repetition	ordinary choice	$\binom{n}{r}$
Unordered, repetition allowed	multisets	$\binom{n+r-1}{r}$

Binomial Coefficients

- Generalized binomial coefficient:

$$\binom{x}{k} = \frac{x(x-1) \cdots (x-k+1)}{k!}, \quad k \geq 0.$$

- Symmetry:

$$\binom{n}{k} = \binom{n}{n-k}.$$

Stars and Bars

□ Nonnegative solutions to $x_1 + \cdots + x_n = r$:

$$\binom{n+r-1}{r}.$$

□ Positive solutions to $x_1 + \cdots + x_u = b$:

$$\binom{b-1}{u-1}.$$

Functions / Balls into Urns

Balls	Urns	Constraint	Count
Labelled	Labelled	Any function	u^b
Labelled	Labelled	Injective	$u(u-1)\cdots(u-b+1)$
Labelled	Labelled	Surjective	$\sum_{\ell=0}^{u-1} (-1)^\ell \binom{u}{\ell} (u-\ell)^b$
Unlabelled	Labelled	Any distribution	$\binom{u+b-1}{b}$
Unlabelled	Labelled	At most one per urn	$\binom{u}{b}$
Unlabelled	Labelled	Onto	$\binom{b-1}{u-1}$

Multinomial Coefficient

□ If $n = n_1 + \cdots + n_k$, then

$$\binom{n}{n_1, \dots, n_k} = \frac{n!}{n_1! \cdots n_k!}.$$

□ It counts ways to split n distinct objects into k distinct groups with prescribed sizes.

Physics Counting Models

- Maxwell-Boltzmann: distinguishable particles, distinct cells, repetition allowed.
- Bose-Einstein: indistinguishable particles, distinct cells, multiple occupancy allowed.
- Fermi-Dirac: indistinguishable particles, distinct cells, at most one particle per cell.

5 Random Variables and Distribution Functions

Random Variable

□ A real-valued random variable is a function

$$X : \Omega \rightarrow \mathbb{R}.$$

Cumulative Distribution Function

□ The CDF of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

- F_X is nondecreasing.
- F_X is right-continuous.
- Boundary conditions:

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow +\infty} F_X(x) = 1.$$

□ Quantile definition:

$$F^{-1}(p) = \inf\{x : F(x) \geq p\}.$$

Discrete Random Variable and PMF

- X is discrete if $X(\Omega)$ is countable.
- PMF:

$$p_X(x) = \mathbb{P}(X = x).$$

□ A function p is a PMF iff

$$p(x) \geq 0, \quad \sum_x p(x) = 1.$$

Continuous Random Variable and PDF

□ X is continuous if there exists f_X such that

$$F_X(x) = \int_{-\infty}^x f_X(y) dy.$$

□ For continuous X ,

$$\mathbb{P}(X = x) = 0.$$

□ A PDF satisfies

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

□ Interval probability:

$$\mathbb{P}(a < X \leq b) = \int_a^b f_X(x) dx.$$

Measure Zero

□ $N \subseteq \mathbb{R}$ has measure zero if for every $\varepsilon > 0$, there are intervals I_k such that

$$N \subseteq \bigcup_{k \geq 1} I_k, \quad \sum_{k \geq 1} \ell(I_k) < \varepsilon.$$

- Countable sets have measure zero.
- Curves in $\mathbb{R}^2$ have area zero.
- PDFs are unique up to measure-zero sets.

6 Expectation, Variance, LOTUS, MGF

Expectation

□ Discrete case:

$$\mathbb{E}(X) = \sum_i x_i p_X(x_i), \quad \sum_i |x_i| p_X(x_i) < \infty.$$

□ Continuous case:

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx, \quad \int_{-\infty}^{\infty} |x| f_X(x) dx < \infty.$$

LOTUS

□ If $Y = \phi(X)$, then in the discrete case

$$\mathbb{E}[\phi(X)] = \sum_i \phi(x_i) p_X(x_i).$$

□ In the continuous case,

$$\mathbb{E}[\phi(X)] = \int_{-\infty}^{\infty} \phi(x) f_X(x) dx.$$

Indicator Function

□ For an event A ,

$$\mathbf{1}_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

□ Indicator expectation:

$$\mathbb{E}(\mathbf{1}_A) = \mathbb{P}(A).$$

Linearity and Jensen

□ Linearity of expectation:

$$\mathbb{E}(\lambda X + \mu Y) = \lambda \mathbb{E}(X) + \mu \mathbb{E}(Y).$$

- No independence is needed for linearity.
- Jensen's inequality: if ϕ is convex,

$$\phi(\mathbb{E}X) \leq \mathbb{E}[\phi(X)].$$

Variance and Standard Deviation

□ Variance:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2], \quad \mu = \mathbb{E}(X).$$

□ Computational form:

$$\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2.$$

□ Standard deviation:

$$\sigma_X = \sqrt{\text{Var}(X)}.$$

□ Scaling:

$$\text{Var}(c) = 0, \quad \text{Var}(cX) = c^2 \text{Var}(X).$$

□ If X, Y are independent,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Moments and Moment Generating Function

□ The k -th ordinary moment is $\mathbb{E}(X^k)$.

□ MGF:

$$m_X(t) = \mathbb{E}(e^{tX}).$$

□ If the MGF exists near 0, it determines the distribution.

□ Moments from MGF:

$$\mathbb{E}(X^k) = m_X^{(k)}(0).$$

□ If X, Y are independent,

$$m_{X+Y}(t) = m_X(t)m_Y(t).$$

7 Discrete Distributions

Bernoulli Distribution

□ $X \sim \text{Bernoulli}(\theta)$ if

$$\mathbb{P}(X = 1) = \theta, \quad \mathbb{P}(X = 0) = 1 - \theta.$$

□ Mean and variance:

$$\mathbb{E}(X) = \theta, \quad \text{Var}(X) = \theta(1 - \theta).$$

Geometric Distribution

□ $X \sim \text{Geometric}(\theta)$ counts the trial number of the first success.

□ PMF:

$$\mathbb{P}(X = x) = (1 - \theta)^{x-1}\theta, \quad x = 1, 2, \dots$$

□ CDF:

$$F_X(x) = 1 - (1 - \theta)^{\lfloor x \rfloor}.$$

□ MGF:

$$m_X(t) = \frac{\theta e^t}{1 - (1 - \theta)e^t}.$$

□ Mean and variance:

$$\mathbb{E}(X) = \frac{1}{\theta}, \quad \text{Var}(X) = \frac{1 - \theta}{\theta^2}.$$

Binomial Distribution

□ $X \sim \text{Binomial}(n, \theta)$ counts successes in n Bernoulli trials.

□ PMF:

$$\mathbb{P}(X = x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}, \quad x = 0, \dots, n.$$

□ MGF:

$$m_X(t) = (1 - \theta + \theta e^t)^n.$$

□ Mean and variance:

$$\mathbb{E}(X) = n\theta, \quad \text{Var}(X) = n\theta(1 - \theta).$$

Negative Binomial Distribution

□ $X \sim \text{NegBinomial}(r, \theta)$ counts trials until the r -th success.

□ If k failures occur before the r -th success, then

$$\mathbb{P}(X = r + k) = \binom{r+k-1}{k} (1 - \theta)^k \theta^r, \quad k = 0, 1, 2, \dots$$

□ $\text{Geometric}(\theta) = \text{NegBinomial}(1, \theta)$.

□ MGF:

$$m_X(t) = \frac{(\theta e^t)^r}{[1 - (1 - \theta)e^t]^r}.$$

□ Mean and variance:

$$\mathbb{E}(X) = \frac{r}{\theta}, \quad \text{Var}(X) = \frac{r(1 - \theta)}{\theta^2}.$$

Hypergeometric Distribution

□ Population size n , with n_1 red and $n - n_1$ non-red. Choose r without replacement.

□ If X is the number of red objects chosen, then

$$\mathbb{P}(X = k) = \frac{\binom{n_1}{k} \binom{n-n_1}{r-k}}{\binom{n}{r}}.$$

□ Support:

$$\max\{0, r - (n - n_1)\} \leq k \leq \min\{r, n_1\}.$$

□ Mean:

$$\mathbb{E}(X) = \frac{rn_1}{n}.$$

□ Variance:

$$\text{Var}(X) = r \cdot \frac{n_1}{n} \cdot \frac{n - n_1}{n} \cdot \frac{n - r}{n - 1}.$$

Poisson Distribution

□ $X \sim \text{Poisson}(\lambda)$ if

$$\mathbb{P}(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

□ MGF:

$$m_X(t) = e^{\lambda(e^t - 1)}.$$

□ Mean and variance:

$$\mathbb{E}(X) = \text{Var}(X) = \lambda.$$

□ If $X \sim \text{Poisson}(\lambda)$, $Y \sim \text{Poisson}(\mu)$, and X, Y are independent, then

$$X + Y \sim \text{Poisson}(\lambda + \mu).$$

Binomial to Poisson Approximation

□ If $X_n \sim \text{Binomial}(n, \theta_n)$ and $n\theta_n = \lambda$, then

$$X_n \Rightarrow \text{Poisson}(\lambda).$$

8 Stochastic Process and Poisson Process

Stochastic Process

□ A stochastic process is a family of random variables

$$(X_t)_{t \in T}.$$

□ For fixed t , X_t is a random variable.

□ For fixed ω , $t \mapsto X_t(\omega)$ is a sample path.

Counting Process

□ A counting process $N = (N_t)_{t \geq 0}$ counts the number of events up to time t .

□ Usually N_t is nondecreasing and right-continuous.

Poisson Process

□ A counting process N_t is a Poisson process with rate λ if $N_0 = 0$.

□ It has independent increments over disjoint intervals.

□ Increment distribution:

$$N_t - N_s \sim \text{Poisson}(\lambda(t - s)), \quad 0 \leq s < t.$$

□ Hence

$$\mathbb{P}(N_t - N_s = k) = \frac{e^{-\lambda(t-s)} [\lambda(t-s)]^k}{k!}.$$

□ No-event probability:

$$\mathbb{P}(N_t = 0) = e^{-\lambda t}.$$

□ Small-time behavior:

$$\lim_{t \rightarrow 0} \frac{1}{t} \mathbb{P}(N_t = 1) = \lambda, \quad \lim_{t \rightarrow 0} \frac{1}{t} \mathbb{P}(N_t \geq 2) = 0.$$

Interarrival Times

□ The first waiting time W_1 in a Poisson process with rate λ satisfies

$$W_1 \sim \text{Exponential}(1/\lambda).$$

□ More generally,

$$W_i \stackrel{\text{iid}}{\sim} \text{Exponential}(1/\lambda).$$

9 Continuous Distributions

Gamma Function

□ Definition:

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx.$$

□ Useful values and recurrence:

$$\Gamma(1) = 1, \quad \Gamma(\alpha + 1) = \alpha \Gamma(\alpha),$$

$$\Gamma(n + 1) = n!, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$

Gamma Distribution

□ $X \sim \text{Gamma}(\alpha, \beta)$ if

$$f_X(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x > 0.$$

□ MGF:

$$m_X(t) = (1 - \beta t)^{-\alpha}, \quad t < 1/\beta.$$

□ Mean and variance:

$$\mathbb{E}(X) = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2.$$

Exponential Distribution

□ $X \sim \text{Exponential}(\beta)$ if

$$f_X(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x > 0.$$

□ CDF:

$$F_X(x) = \begin{cases} 1 - e^{-x/\beta}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

□ MGF:

$$m_X(t) = \frac{1}{1 - \beta t}.$$

□ Mean and variance:

$$\mathbb{E}(X) = \beta, \quad \text{Var}(X) = \beta^2.$$

Chi-Squared Distribution

□ $X \sim \chi_k^2$ iff

$$X \sim \text{Gamma}\left(\frac{k}{2}, 2\right).$$

□ PDF:

$$f_X(x) = \frac{1}{\Gamma(k/2)2^{k/2}} x^{k/2-1} e^{-x/2}, \quad x > 0.$$

□ Mean and variance:

$$\mathbb{E}(X) = k, \quad \text{Var}(X) = 2k.$$

Normal Distribution

□ $X \sim N(\mu, \sigma^2)$ if

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}.$$

□ MGF:

$$m_X(t) = e^{\mu t + \sigma^2 t^2/2}.$$

□ Mean and variance:

$$\mathbb{E}(X) = \mu, \quad \text{Var}(X) = \sigma^2.$$

□ Standardization:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

□ Standard normal CDF:

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

□ 68-95-99.7 rule:

$$\mathbb{P}(|X - \mu| < \sigma) \approx 0.68,$$

$$\mathbb{P}(|X - \mu| < 2\sigma) \approx 0.95,$$

$$\mathbb{P}(|X - \mu| < 3\sigma) \approx 0.997.$$

Transformation of Continuous Random Variables

□ If $Y = g(X)$, where g is strictly monotone and C^1 , then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

□ If g is not one-to-one, use the CDF method:

$$F_Y(y) = \mathbb{P}(g(X) \leq y),$$

then differentiate.

Weibull Distribution

□ $X \sim \text{Weibull}(\alpha, \beta)$ if

$$f_X(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, \quad x > 0.$$

□ Mean:

$$\mathbb{E}(X) = \alpha^{-1/\beta} \Gamma\left(1 + \frac{1}{\beta}\right).$$

□ Variance:

$$\text{Var}(X) = \alpha^{-2/\beta} \left[\Gamma\left(1 + \frac{2}{\beta}\right) - \Gamma\left(1 + \frac{1}{\beta}\right)^2 \right].$$

□ Reliability:

$$R(t) = \mathbb{P}(X > t) = e^{-\alpha t^\beta}.$$

□ Hazard rate:

$$\rho(t) = \frac{f(t)}{R(t)} = \alpha\beta t^{\beta-1}.$$

□ Hazard interpretation:

$$\begin{array}{ll} \beta > 1 & \text{increasing hazard,} \\ \beta = 1 & \text{constant hazard,} \\ 0 < \beta < 1 & \text{decreasing hazard.} \end{array}$$

□ $\text{Exponential}(1/\alpha) = \text{Weibull}(\alpha, 1)$.

10 Reliability Systems

Reliability Function and Hazard Rate

□ For failure time X ,

$$R(t) = \mathbb{P}(X > t) = 1 - F(t).$$

□ Hazard rate:

$$\rho(t) = \frac{f(t)}{R(t)}.$$

□ Reliability from hazard:

$$R(t) = \exp\left[-\int_0^t \rho(s) ds\right].$$

Series and Parallel Systems

□ Series system: fails whenever any component fails.

$$R_{\text{series}}(t) = \prod_{i=1}^k R_i(t).$$

□ Parallel system: fails only when all components fail.

$$R_{\text{parallel}}(t) = 1 - \prod_{i=1}^k (1 - R_i(t)).$$

11 Inequalities, LLN, and CLT

Markov Inequality

□ If $\mathbb{E}|X| < \infty$, then

$$\mathbb{P}(|X| \geq \varepsilon) \leq \frac{\mathbb{E}|X|}{\varepsilon}.$$

□ If $X \geq 0$, then

$$\mathbb{P}(X \geq \varepsilon) \leq \frac{\mathbb{E}(X)}{\varepsilon}.$$

□ Survival-function expectation for $X \geq 0$:

$$\mathbb{E}(X) = \int_0^{\infty} \mathbb{P}(X > x) dx.$$

□ Integer-valued case:

$$\mathbb{E}(X) = \sum_{k \geq 0} \mathbb{P}(X > k) = \sum_{k \geq 1} \mathbb{P}(X \geq k).$$

Chebyshev and Cantelli Inequalities

□ Chebyshev inequality:

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

□ Equivalent form:

$$\mathbb{P}(|X - \mu| > a) \leq \frac{\sigma^2}{a^2}.$$

□ Cantelli inequality:

$$\mathbb{P}(X - \mu \geq \lambda) \leq \frac{\sigma^2}{\sigma^2 + \lambda^2},$$

$$\mathbb{P}(X - \mu \leq -\lambda) \leq \frac{\sigma^2}{\sigma^2 + \lambda^2}.$$

Weak Law of Large Numbers

□ If $X_1, X_2, \dots$ are iid with mean μ and variance $\sigma^2 < \infty$, then

$$\mathbb{P}\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

□ Meaning: sample average converges in probability to the mean.

De Moivre-Laplace Theorem and CLT

□ If $S_n \sim \text{Binomial}(n, \theta)$, then

$$\frac{S_n - n\theta}{\sqrt{n\theta(1-\theta)}} \Rightarrow N(0, 1).$$

□ Central Limit Theorem: if $X_1, X_2, \dots$ are iid with mean μ and variance $\sigma^2 < \infty$, then

$$\frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \Rightarrow N(0, 1).$$

Continuity Correction

□ For $X \sim \text{Binomial}(n, \theta)$,

$$\mathbb{P}(a \leq X \leq b) \approx \Phi\left(\frac{b + \frac{1}{2} - n\theta}{\sqrt{n\theta(1-\theta)}}\right) - \Phi\left(\frac{a - \frac{1}{2} - n\theta}{\sqrt{n\theta(1-\theta)}}\right).$$

□ Rule of thumb:

$$n\theta > 5, \quad n(1-\theta) > 5.$$

Mean vs Median

□ Median minimizes absolute error:

$$m = \arg \min_z \mathbb{E}|X - z|.$$

□ Mean minimizes squared error:

$$\mu = \arg \min_z \mathbb{E}[(X - z)^2].$$

□ Mean-median bound:

$$|\mu - m| \leq \sigma.$$

12 Multivariate Random Variables

Discrete Random Vector and Joint PMF

□ A discrete random vector is

$$X = (X_1, \dots, X_n) : \Omega \rightarrow \mathbb{R}^n.$$

□ Joint PMF:

$$p_{X_1, \dots, X_n}(x_1, \dots, x_n) = \mathbb{P}(X_1 = x_1, \dots, X_n = x_n).$$

□ A joint PMF satisfies

$$p(x_1, \dots, x_n) \geq 0, \quad \sum_{(x_1, \dots, x_n)} p(x_1, \dots, x_n) = 1.$$

Marginal PMF and Conditional PMF

□ Bivariate marginals:

$$p_X(x) = \sum_y p_{XY}(x, y), \quad p_Y(y) = \sum_x p_{XY}(x, y).$$

□ Conditional PMF:

$$p_{X|Y}(x | y) = \frac{p_{XY}(x, y)}{p_Y(y)}, \quad p_Y(y) > 0.$$

Independence of Discrete Random Variables

□ $X_1, \dots, X_n$ are independent iff

$$p_{X_1, \dots, X_n}(x_1, \dots, x_n) = p_{X_1}(x_1) \cdots p_{X_n}(x_n).$$

Continuous Random Vector and Joint Density

□ Joint density f_X satisfies

$$f_X(x) \geq 0, \quad \int_{\mathbb{R}^n} f_X(x) dx = 1.$$

□ Probability over a region:

$$\mathbb{P}(X \in D) = \int_D f_X(x) dx.$$

□ Joint CDF:

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n).$$

□ If density exists,

$$f_{X_1, \dots, X_n} = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{X_1, \dots, X_n} \quad \text{a.e.}$$

Marginal Density

□ Bivariate marginals:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx.$$

Independence of General Random Variables

□ General criterion:

$$F_{X_1, \dots, X_n}(x_1, \dots, x_n) = F_{X_1}(x_1) \cdots F_{X_n}(x_n).$$

□ Continuous case:

$$f_{X_1, \dots, X_n} = f_{X_1} \cdots f_{X_n} \quad \text{a.e.}$$

□ Discrete case:

$$p_{X_1, \dots, X_n} = p_{X_1} \cdots p_{X_n}.$$

Expectation of Random Vectors

□ Mean vector:

$$\mathbb{E}(X) = \begin{bmatrix} \mathbb{E}(X_1) \\ \vdots \\ \mathbb{E}(X_n) \end{bmatrix}.$$

□ Discrete LOTUS:

$$\mathbb{E}[\phi(X)] = \sum_{x \in \mathbb{R}^n} \phi(x) p_X(x).$$

□ Continuous coordinate mean:

$$\mathbb{E}(X_k) = \int_{\mathbb{R}^n} x_k f_{X_1, \dots, X_n}(x_1, \dots, x_n) dx_1 \cdots dx_n.$$

13 Covariance and Correlation

Covariance

Definition:

Cov(X, Y) = E[(X - μ_X)(Y - μ_Y)].

Computational form:

Cov(X, Y) = E(XY) - E(X)E(Y).

Properties:

Cov(X, Y) = Cov(Y, X), Cov(X, X) = Var(X).

Linearity:

Cov(αX + βY, Z) = α Cov(X, Z) + β Cov(Y, Z).

Cauchy-Schwarz:

|Cov(X, Y)| ≤ √Var(X) Var(Y).

- Independence implies zero covariance.
- Zero covariance does **not** imply independence.

Covariance Matrix

For X = (X1, ..., Xn), the covariance matrix is

CX = (cij), cij = Cov(Xi, Xj).

CX is symmetric:

CX = CTX.

CX is positive semidefinite:

vT CX v ≥ 0.

Linear transformation:

CAX = ACX AT.

Correlation Coefficient

Definition:

ρXY = Cov(X, Y) / √Var(X) Var(Y) = Cov(X, Y) / σX σY.

Bound:

|ρXY| ≤ 1.

Perfect correlation:

|ρXY| = 1 ⇔ Y = β0 + β1X a.s.

ρ = 0 means no linear correlation, not necessarily independence.

14 Conditional Distribution, Conditional Expectation, Regression

Conditional Density

For joint density fXY,

fX|Y(x | y) = fXY(x, y) / fY(y), fY(y) > 0.

Similarly,

fY|X(y | x) = fXY(x, y) / fX(x).

Conditional Expectation

Discrete case:

E(X | Y = y) = ∑x x pX|Y(x | y).

Continuous case:

E(X | Y = y) = ∫x fX|Y(x | y) dx.

E(X | Y) is a random variable obtained by plugging in Y.

Total Expectation and Total Variance

Law of total expectation:

E[E(X | Y)] = E(X).

Conditional variance:

Var(X | Y) = E(X² | Y) - [E(X | Y)]².

Law of total variance:

Var(X) = E[Var(X | Y)] + Var(E(X | Y)).

Properties of Conditional Expectation

Constant property:

E(α | Y) = α.

Linearity:

E(αX + βZ | Y) = αE(X | Y) + βE(Z | Y).

Positivity: if X ≥ 0 a.s., then E(X | Y) ≥ 0.

Independence: if X and Y are independent, then

E(X | Y) = E(X).

Pull-out property:

E[Xg(Y) | Y] = g(Y)E(X | Y).

Tower property:

E[E(X | Y, Z) | Y] = E(X | Y).

Regression and Best Predictor

Regression curve of X on Y:

y ↦ E(X | Y = y).

Regression curve of Y on X:

x ↦ E(Y | X = x).

Best mean-square predictor:

E(Y | X) = arg min_h E[(Y - h(X))²].

15 Key Distinctions and Common Traps

Probability vs Expectation

- Expected number of successes is not the same as probability of at least one success.
- Correct formula:

P(at least one success) = 1 - (1 - p)ⁿ.

Approximation:

1 - (1 - p)ⁿ ≈ np

only when p is small and np is moderate.

Distribution Distinctions

Distribution	Fixed	Random
Binomial	number of trials n	number of successes
Geometric	first success target	waiting time until first success
Negative binomial	r-th success target	waiting time / failures before r-th success

Distribution	Replacement?	Independence?
Binomial	with replacement / independent trials	yes
Hypergeometric	without replacement	no

PMF vs PDF

□ PMF gives actual point probabilities:

$$p_X(x) = \mathbb{P}(X = x).$$

□ PDF is density; for continuous variables,

$$\mathbb{P}(X = x) = 0.$$

□ Continuous probabilities are areas:

$$\mathbb{P}(a < X < b) = \int_a^b f_X(x) \, dx.$$

Independence vs Uncorrelated

□ Independence implies zero covariance:

$$X, Y \text{ independent} \Rightarrow \text{Cov}(X, Y) = 0.$$

□ Zero covariance does not imply independence:

$$\text{Cov}(X, Y) = 0 \not\Rightarrow X, Y \text{ independent}.$$

Pairwise vs Mutual Independence

□ Pairwise independence checks pairs only.

□ Mutual independence checks all intersections/subcollections.

LLN vs CLT

□ LLN: sample average converges to the mean.

$$\bar{X}_n \rightarrow \mu.$$

□ CLT: normalized fluctuation becomes normal.

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \Rightarrow N(0, 1).$$